CLASS 9 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 3 & 4: **The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The correct name of the covalent compound SF_6 is: 1

(a) disulfur hexafluoride **(b)** sulfur fluoride
(c) sulfur hexafluoride **(d)** sulfur hexafluorine

2. A molecule is best defined as: 1

(a) an electrically neutral entity of more than one atom capable of independent existence
(b) only a particle of a compound
(c) a charged particle of an element
(d) a single atom of any element

3. In the formation of NaCl, sodium: 1

(a) loses two electrons **(b)** shares one electron
(c) gains one electron to form Na^- **(d)** loses one electron to form Na^+

4. Which of the following is a typical property of ionic compounds? 1

(a) soft and flammable **(b)** low melting point
(c) poor solubility in water always **(d)** high melting and boiling points

5. A formula unit mass is used for: 1

(a) ionic compounds **(b)** covalent compounds
(c) elements only **(d)** gases only

6. In the gold foil experiment most α-particles passed straight through. This showed that: 1

(a) most of the atom is empty space **(b)** gold is radioactive
(c) atoms are solid spheres **(d)** α-particles are neutral

7. Bohr's model proposes that electrons: 1

(a) move randomly with changing energy
(b) revolve in fixed shells without losing energy
(c) have no energy
(d) are fixed at the nucleus

8. The atomic number of an element is equal to the number of: 1

(a) protons in its nucleus **(b)** valence electrons
(c) neutrons **(d)** nucleons

9. The isotope used in the treatment of goitre is: 1

(a) cobalt-60 **(b)** uranium-235 **(c)** iodine-131 **(d)** carbon-14

10. An atom with an octet in its valence shell is generally: 1

(a) highly reactive **(b)** negatively charged always

(c) radioactive **(d)** unreactive and stable

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Differentiate between molecular mass and formula unit mass. 2

12. Give two reasons why ionic compounds have high melting points. 2

13. Why do isotopes of an element have similar chemical properties? 2

14. Define isobars with an example. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Why do noble gases exist as single atoms? Give two examples. 3

16. Why are noble gases chemically unreactive? Give configurations of neon and argon. 3

17. What are the energy changes when atoms combine? Why are molecules more stable than separate atoms? 3

OR

Why is hydrogen written first without a prefix in H_2S? Name H_2O and NH_3 using common names. 3

18. Two atoms have 11 protons each, one with 12 and the other with 13 neutrons. Compare their atomic numbers and mass numbers. Are they the same element? 3

OR

Why is it said that electrons do not move in well-defined orbits as in Bohr's model? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What is a molecule? 5

(ii) Distinguish between a molecule of an element and of a compound with two examples each.

(iii) Why does helium exist as single atoms?

OR

(i) Show that the combining ratio of carbon and oxygen in CO_2 is 3 : 8. 5

(ii) How much carbon is in 22 g of CO_2?

(iii) Find the mass of oxygen needed to burn 6 g of carbon completely.

20. (i) Why were symbols introduced for elements? 5

(ii) State the norms for writing symbols.

(iii) Give the symbols of iron, mercury, sodium and potassium and their origins.

OR

(i) Explain why electrons were thought to be present in every atom. 5

(ii) Why is the atom electrically neutral?

(iii) Calculate the number of subatomic particles in an atom of aluminium (Z = 13, A = 27) and write its configuration and valency.

END OF QUESTION PAPER Total: 40 Marks